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Highlights

- Strength requirement is considered in additive manufacturing scheduling problem
- Considering strength requirements may help reduce makespan and total tardiness
- ε-constraint and non-dominated sorting genetic algorithm-II (NSGA-II) are developed
- New decoding mechanisms integrated into the NSGA-II improve search performance

Including mechanical requirements in a bi-objective nesting and scheduling model for additive manufacturing

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Abstract

Following the increasing relevance of Additive Manufacturing (AM) as Manufacturing-as-a-service (Maas), the AM scheduling (and related nesting) problem has been increasingly investigated. Due to their business nature, Maas companies are interested in minimizing both the makespan and the total tardiness; however, most of the literature focuses only on one of them. This work fills this gap proposing a mixed-integer linear programming (MILP) model that minimizes both makespan and total tardiness. In doing so, for the first time in the literature, considerations on parts' strength are included. During nesting procedures, indeed, parts can be oriented in different ways, with this choice affecting not only the total processing time (as considered by the literature) but also the strength achievable: if this is lower than what planned, parts might fail unexpectedly with detrimental consequences. Thus, this work ensures that parts are produced with the required strength. In doing so, we focus on a parallel unrelated AM batch scheduling problem for metallic parts.

Considering the multi-objective and NP-hard nature of the problem, an ε-constraint algorithm and a non-dominated sorting genetic algorithm-II (NSGA-II) are developed to solve the problem. Four different problem-specific decoding mechanisms are integrated into the NSGA-II to improve its search capability and solution-building performance. Their performances are evaluated through computational experiments, showing that the integrated mechanisms improve the performance of the NSGA-II. Finally, through numerical instances and analysis of the super Pareto front, we derive managerial insights on the impact of strength requirements and machines' number and features on the objectives.

Keywords: Scheduling, additive manufacturing (AM), 2D nesting, technological constraints, NSGA-II

1. Introduction

In the last years, Additive Manufacturing (AM), also known as 3D printing, has emerged as a disruptive manufacturing technology (Westerweel et al., 2018; Peron, 2024), attracting growing interest among practitioners. According to a market report (SphericalInsights, 2024), the AM market will reach $143.3 billion by 2033 from $19.97 billion in 2023, with an estimated annual growth rate of 21.8%. Among the sectors most interested in adopting AM, aerospace, automotive, healthcare, and defense sectors lead the way, with a strong focus on AM for metallic parts and components. Specifically, these sectors are interested in exploiting two of the main benefits of AM: (i) the design flexibility achievable, which allows manufacturing parts and components of complex shapes (Aloui & Hadj-Hamou, 2021), and (ii) the negligible tooling and setup times required, which allow for on-demand production (Kucukkoc, 2019).

Following this increasing interest toward AM, the literature has investigated its implications for business performance in an attempt to understand the most convenient business model to adopt (Jiang et al., 2017; Sgarbossa et al., 2021). The literature concurs that the main drawback of AM is the high investment costs necessary to purchase AM machines (Altekin & Bukchin, 2022), especially when metallic parts and components are needed. Therefore, a business model in which the AM machine usage rate and efficiency are maximized is necessary to ensure and enhance the economic competitiveness of AM (Yu et al., 2022). This is the case of the Manufacturing-as-a-service (Maas) business model, where several customers outsource their production to a limited number of contractors (i.e., manufacturers owning the AM machines) that are hence able to maximize AM machine usage rate and efficiency. These manufacturers, however, must guarantee that the customers' requirements in terms of the required delivery dates and quality are met; otherwise, penalty costs arise. Consequently, with the aims of maximizing the AM machine usage rate and ensuring that customers' requirements are met, scheduling the production efficiently becomes an important issue for Maas, especially considering that more parts can be simultaneously produced by the same machine (Kucukkoc, 2019).

The scheduling problem for AM machines has received increasing attention over the last decade (Oh et al., 2020). In AM scheduling, manufacturers collect orders of parts and components from customers, where each part is associated with characteristics such as sizes, quality requirements (particularly in terms of strength), and due dates. Then, the parts and components are grouped into different job batches, which are then allocated to different AM machines according to their features (e.g., dimensions) (Li et al., 2017). Moreover, the scheduling problem is strongly interconnected with the nesting (or packing) problem, which concerns the placement of parts in the build platform, and thus highly affects the total processing time (Oh et al., 2020). The decisions include part placement and orientation (i.e., the angle with respect to the building platform and rotation around such an angle, cf. Section 3 for more details). Notably, the latter is constrained since it affects the strength of the parts and components, which is a stringent and critical requirement in many applications. If parts are produced with a strength lower than what was planned during the design phase, these would not be able to withhold the loads as required. Hence, these parts would fail unexpectedly and prematurely, with detrimental consequences for safety and costs (Siemens, 2020).

However, as will be better discussed in Section 2, the current literature on this topic is characterized by some main limitations. First, despite the interrelations between scheduling and nesting, either nesting is not considered, or these two problems are mostly tackled hierarchically in a two-step approach. The parts are first grouped in batches and their orientation is set (nesting). Then, batch production is assigned and sequenced among different AM machines (scheduling) (Chergui et al., 2018). However, as reported by Kucukkoc (2019), this subsequent approach does not lead to optimal solutions, and nesting and scheduling problems should therefore be solved jointly. Second, most existing works focus only on makespan minimization (i.e., on the minimization of the latest completion time of parts on all machines). However, considering the nature of Maas, where manufacturers must guarantee product delivery before a due date, tardiness (i.e., the delay in completing the production after the planned due date of parts) should also be considered and minimized. Third, when considering the nesting problem, the existing literature completely neglects the fact that decisions related to the orientation of parts and components must follow the design requirements in terms of achievable strength to avoid unexpected and premature failures.

In light of the above-mentioned limitations, this work aims to overcome these three literature gaps. Specifically, we propose a bi-objective model that minimizes both the makespan and the total tardiness. More in detail, the model jointly considers scheduling and nesting problems, where the latter is considered as a 2D nesting problem. This is due to the relevance that metallic parts have (and will have) in AM adoption; hence, we focus on the most common manufacturing technique to produce metallic AM parts, i.e., the Selective Laser Melting (SLM) process, where no stacks are permitted (and hence, 3D nesting is not an option) (Yu et al., 2022). Finally, for the first time in the literature, we directly link the orientation of parts to their required strength. Due to the strongly NP-hard nature of this problem, we propose a non-dominated sorting genetic algorithm-II (NSGA-II) to solve medium- and large-size instances. For small-size instances, we optimally solve the model by applying the ε-constraint method. Finally, an in-depth analysis of the super Pareto front is carried out to derive managerial insights.

The remainder of this paper is structured as follows. Section 2 reviews the literature on the topic, highlighting the novelties of this work. Section 3 describes the problem under consideration and the assumptions made, followed in Section 4 by the mathematical model. Section 5 reports the solution approaches (the ε-constraint method and NSGA-II). Section 6, then, reports the computational experiment and the managerial insights. Finally, Section 7 reports conclusions and future research directions.

2. Literature review

As reported by Li et al. (2017), the AM machine scheduling problem differs from the traditional batch scheduling problem with arbitrary job sizes (Fanjul-Peyro et al., 2017). Furthermore, it is more complex to solve than the traditional single-batch machine scheduling problem (which is an NP-hard problem). Consequently, due to the differences with the traditional single-batch machine scheduling problem and the increased complexity, new models, methods, and techniques have been proposed in recent years. The first work to deal with the AM machine scheduling problem is that of Kucukkoc et al. (2016), which, despite its limitations (e.g., no computational experiments were performed), can be seen as the pioneering work in the field, following which academics and researchers started to develop new models and solving approaches (Pinto et al., 2024).

Initially, the literature focused on single-objective models, where AM machine scheduling was solved according to a single-objective function (mostly either minimization of the makespan, minimization of the costs, maximization of the profit, or minimization of the tardiness) (Oh et al., 2020). An example of this includes the work of Kucukkoc (2019), where the author developed a MILP model to minimize the makespan, considering different machine configurations (i.e., single machine, parallel identical machines, and parallel non-identical machines). Alicastro et al. (2021), then, extended the work of Kucukkoc (2019) by developing a reinforcement learning iterated local search meta-heuristic to provide heuristically good solutions at the cost of low computational expenses. However, these works, as well as the other initial works on AM machine scheduling problems, did not include the nesting problem, which (as discussed above) should be considered (Oh et al., 2020).

Initially, the nesting problem was considered separately from the scheduling problem, in a two-step approach: first, parts are grouped into batches (or jobs), and their orientation is set (nesting problem); then, batch (or job) production is assigned and sequenced among different AM machines (scheduling problem). This is the example of Zhang et al. (2016) and Chergui et al. (2018), who adopted a "parallel nesting" algorithm (Zhang et al. 2016) and a heuristic (Chergui et al., 2018) to solve the nesting problem. However, nesting and scheduling problems must be considered jointly to avoid suboptimal solutions (Kucukkoc, 2019). Some of the first researchers to do this were Li et al. (2020), who developed a MIP model to minimize the makespan of a single-batch processing machine, jointly considering the scheduling problem and the 2D nesting problem. This work was further extended by Oh et al. (2018), who also developed a mathematical model to minimize the makespan that jointly considered the 2D nesting and scheduling problems, but they considered multiple parallel machines. Then, Zipfel et al. (2025) proposed a new branch-and-cut approach to minimize the makespan on parallel unrelated AM machines considering 2D rectangular packing of parts.

However, when considering the 2D nesting problem, these works, as well as other similar works in the literature (e.g., Li et al., 2019; Mao et al., 2024) are characterized by one main limitation, which relates to the orientation of parts with respect to the building platform. Parts can be oriented in many ways (cf. Section 3 for more details), and this decision impacts not only the number of nested parts and the batch processing time, but also the parts' strength achievable, which cannot be lower than what was planned during the design phase (cf. above) (Taufik & Jain, 2013). However, in the above-mentioned works, the orientation is always predetermined: while, on the one hand, this *"simplifies the problem,"* on the other hand, it *"restricts the potential to find better schedules by adopting different orientation strategies"* (Yu et al., 2022). Therefore, in light of further improving the solutions to the scheduling and nesting problems, the orientation of the parts must not be predetermined but must represent an outcome of the analysis. Only a few authors have done this (Griffiths et al., 2019; Oh et al., 2019; Che et al., 2021). Oh et al. (2019), for example, developed a model to minimize the makespan and they allowed the part to be printed according to two main orientations. A similar work is that of Che et al. (2019), who again developed a model to minimize the makespan but allowed more than two possible orientations (these represent an input of the model). A step further in terms of including part orientations in the AM scheduling problem was that of Griffiths et al. (2019), where the authors adopted a Tabu search to automatically explore a solution space of 65,160 different orientations. However, in all these works, the choice of the orientation is based solely on parameters that affect the objective function (either minimization of the makespan, total cost, or total tardiness), such as geometry, height, total support structure volume, and area of the resulting part. Nevertheless, they do not consider the effect of the parts' orientation on the strength of the parts. As discussed above, this represents a big limitation, especially for metallic parts that are required to withhold external loads: if parts are printed with an orientation that reduces their strength, then this would result in unexpected and premature failures of these parts, with detrimental consequences.

Very recently, then, the literature has started moving its focus from single-objective models to multi-objective models, which are summarized in Table 1. However, the bi-objective models are also characterized by the same limitations that have just been highlighted for the single-objective ones. For example, we can see from Table 1. that the focus on the nesting problem is even more limited in the case of multi-objective models, with only one paper considering the nesting problem (Tafakkori et al., 2022). Specifically, they considered the nesting problem (a 2D nesting problem) jointly with the scheduling problem by proposing a novel integrated framework. However, again, the decisions on parts' orientation were not linked to the strength of the parts. Therefore, this work aims to fill the existing gap by extending the current body of literature on multi-objective models that jointly consider nesting and scheduling problems, overcoming the lack of a research focus on the link between nesting and parts' strength. To do so, we have developed a bi-objective model that minimizes both the makespan and the total tardiness, while jointly considering the scheduling and nesting problems. Furthermore, the choice concerning parts' orientation is linked not only to the effects that this has on the objective functions, but also to the effects that this has on the parts' strength.

Table 1. Overview of the multi-objective scheduling model in the AM process

Authors (year)	Objective	Manufacturing technology	AM machine	Nesting	Joint nesting and scheduling	Strength	Mathematical model	Solving approach
Ransikarbum et al. (2017)	MIN operating cost, load balance, total tardiness, and total number of unprinted parts	Fused deposition modeling (FDM)	Multiple non-identical	No	N/A	No	MILP	Meta-heuristic
Fera et al. (2018)	MIN cost and tardiness	SLM	Single	No	N/A	No	MILP	Meta-heuristic
Fera et al. (2020)	MIN cost and tardiness	SLM	Single	No	N/A	No	MILP	Meta-heuristic
Ransikarbum et al. (2020)	MIN operating cost, load balance, total tardiness, and total number of unprinted parts	Fused deposition modeling (FDM)	Multiple non-identical	No	N/A	No	MILP	Meta-heuristic
Tavakkoli-Moghaddam et al. (2020)	MIN makespan and total tardiness	SLM	Multiple non-identical	No	N/A	No	MILP	ε-constraint method
Rohaninejad et al. (2021)	MIN makespan and total tardiness	SLM	Multiple non-identical	No	N/A	No	MILP	Meta-heuristic
Altekin & Bukchin (2022)	MIN cost and makespan	Direct metal Laser Sintering/SLM	Single, Multiple identical, Multiple non-identical	No	N/A	No	MILP	Optimally solved
Tafakkori et al. (2022)	MAX profit, MIN energy utilization, and goodwill losses	Not mentioned	Multiple non-identical	2D	Yes	No	MILP	Three meta-heuristics
Our work	MIN makespan and total tardiness	SLM	Multiple non-identical	2D	Yes	Yes	MILP	Meta-heuristic/ ε-constraint method

3. Problem description and assumptions

In this section, we describe the problem and its main assumptions. As discussed in the Introduction, we aim to solve the AM scheduling problem in one of the most commonly used AM technologies for metallic parts, i.e., SLM. For the sake of brevity and given the high number of papers describing it, we assume that the reader is already familiar with how the SLM process works. If not, the reader can refer to the literature (e.g., Li et al., 2017).

In a general AM scheduling problem, manufacturers receive orders from customers to produce different AM parts, which are characterized by different sizes, release times, due dates, and required strengths. These parts must be allocated to different batches (also known as "jobs"), which are produced in one or more AM machines (which can be identical or non-identical) at different times. Notably, non-identical machines are characterized by different sizes of the building chamber (i.e., different heights, widths, and lengths), as well as by different processing times, including the setup time, volumetric printing time, and recoater time. For the sake of clarity regarding the setup time, we refer to it as the time required to clean the AM machine and initialize the operations required to start production (e.g., AM machine parameter settings, filling the machine with powder materials, filling of the protective atmosphere, etc.); regarding the volumetric printing time, we refer to it as the time required by the laser beam to densify a unitary volume (e.g., 1 cm^3) of powder according to the required cross-section; regarding the recoater time, we refer to it as the time required by the recoater blade to spread one new powder layer in the printing area; this procedure must be repeated for each required layer until the highest part of the job is completed. For more details, the reader can refer to the literature (e.g., Kucukkoc, 2019).

Notably, when forming a batch, some constraints must be considered. First, orders (and hence, parts) arrive at different times (i.e., they have different release times); consequently, the production of a batch cannot be earlier than the latest release time of the parts. Moreover, parts are characterized by due dates; hence, they should ideally be produced before such dates. Then, the number of parts that can be batched and printed in a given AM machine is limited by the capacity of such a machine in terms of the height, width, and length of the building chamber. Thus, the placement of the parts within the build chamber (i.e., nesting) needs to ensure that such geometrical constraints are respected, and that the parts are neither overlapping nor stacking (SLM does not allow building one part on top of the other). Notably, the size of each part is not constant but depends on the orientation of the parts, i.e., the production angle (the angle between the major axis of the part and the building platform) and the rotation of the part around such an angle. Different production angles and part rotations lead to different part sizes (i.e., different heights, widths, and lengths), as well as different volumes of support structures, hence affecting the processing time. Moreover, the production angle affects the parts' strength: from the material science literature, it emerges that parts manufactured with their major axis inclined at 45° with respect to the building platform allows to achieve the maximum strength, while parts manufactured with an angle of 0° and 90° with respect to the building platform result in lower strength (7% and 15% lower, respectively) (Pan et al., 2020). No effect of the parts' rotation on their strength is instead reported. To better clarify the concepts of production angles and part rotations, Figure 1a reports an example of three different production angles (i.e., the major axis of the parts is inclined at 0°, 45°, and 90° with respect to the building platform), while Figure 1b-d shows three different rotations of the part characterized by a production angle equal to 0°.

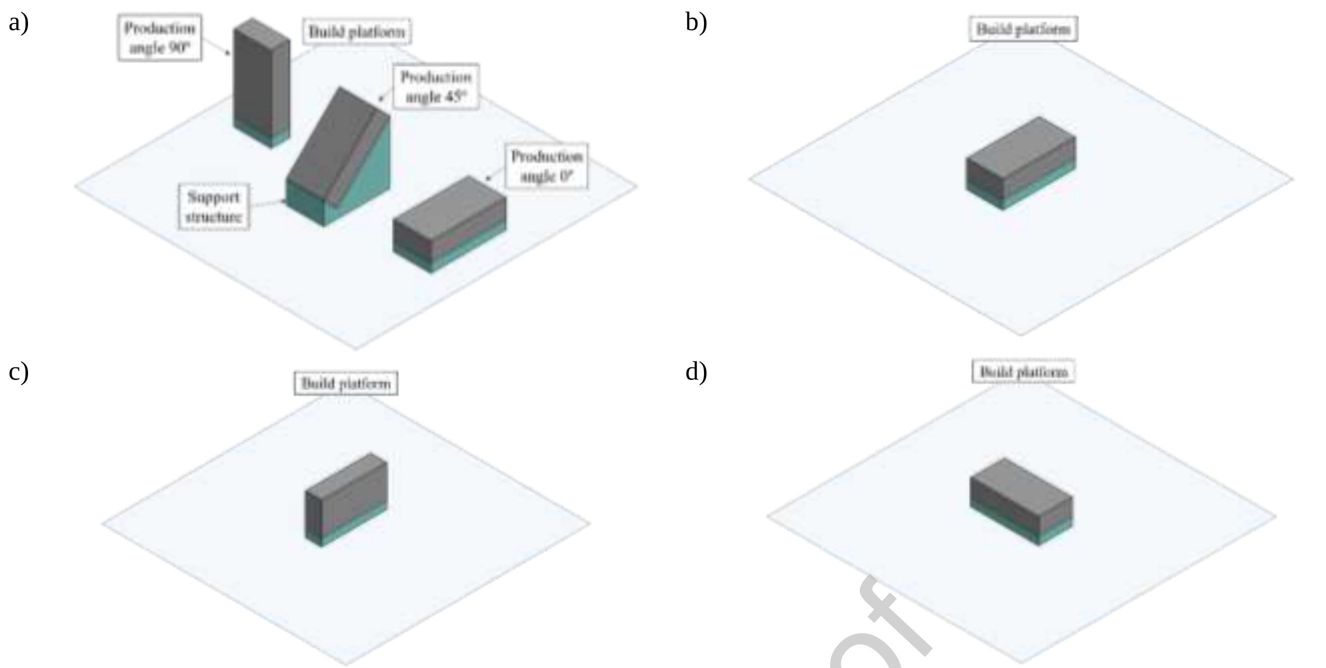

Figure 1. Part with different production angles (a); different rotations of the part characterized by a production angle equal to 0° (b-d).

Considering now the specifics of this study, we focus on an AM machine scheduling problem where a set of parts I can be produced on a set of parallel non-identical AM machines M. Each part $i \in I$ is characterized by a certain volume v_i and by a certain required strength ratio SRP_i. This is defined as the ratio between the part's required strength (information known from the part's designer) and the maximum strength achievable (information known from the part's manufacturer). As an example, parts with aesthetic functions do not require to have the maximum strength achievable, and hence they require an $SRP < 1$; on the contrary, parts with load-bearing functions require the maximum strength achievable, and hence an $SRP = 1$ or close to it. Notably, in this work with the term "strength" we refer to the yield strength, as this is the common strength used during the design of parts and components (EN 1993-1-1; ISO 898-1; ASTM A992/ASTM A992M). However, the methodology herein developed is agnostic for any other strength of interest.

Parts can hence be produced with a production angle $g \in G$ able to satisfy the required strength SRP_i (three production angles are considered in this work, i.e., 0°, 45°, and 90°); then, around each feasible production angle, parts can assume three rotations $n \in N$. These do not influence the strength achievable, but together with the production angle, they influence the height, width, and length of the part (h_{ing}, w_{ing}, and l_{ing}, respectively), and the volume of the support structure s_{ing}.

Moreover, each part i has a specific release time r_i (which represents the earliest time when a part can be produced) and due date dd_i (which represents the latest time when a part should ideally be produced). Each machine $m \in M$, then, is characterized by a different size of the building chamber in terms of height, width, and length (H_m, W_m, and L_m, respectively). Moreover, each machine $m \in M$ is characterized by a different setup time SET_m, volumetric printing time VT_m, and recoater time HT_m. For each batch $j \in J$ on a machine $m \in M$, then, the assigned parts, in addition to respect the capacity of the machine $m \in M$, are required to be packed without any overlap. To avoid the parts' overlapping, we consider a 2D nesting problem (a.k.a. a 2D bin-packing problem), and the parts are considered through their projected rectangular bounding boxes. Finally, the processing time p_{jm} of each batch $j \in J$ on the machine $m \in M$ is determined by the sum of the scanning time, recoater time, and setup time of the machine. The objective is to minimize the makespan (C_{max}, which is the latest completion time on all machines) and the total tardiness of the parts, $\sum_i T_i$. Other assumptions are that (i) only one batch can be processed at a time, (ii) the layer thickness is the same for all parts (60 μm), and (iii) the parts are available at their release date.

4. Mathematical model

Before describing the mathematical model developed (which is a mixed-integer linear program, MILP), we provide the list of notations.

Indices

i, k	part index, $i, k \in I$
j	batch index, $j \in J$
m	machine index, $m \in M$
g	angle index, $g \in G$
n	rotation index, $n \in N$

Parameters

lt	layer thickness [cm]
v_i	volume of part i [cm^3]
r_i	release time of part i [h]
dd_i	due date of part i [h]
$h_{ing}, w_{ing}, l_{ing}, s_{ing}$	height, width, length, and volume of the support structure needed for part i in the rotation n at angle g [cm]
H_m, W_m, L_m	maximum height, width, and length supported by machine m [cm]
VT_m	time required by machine m to densify (print) a unitary volume of powder [h/cm^3]
HT_m	time spent to spread one new powder layer on machine m [h]
SET_m	setup time of machine m [h]
ψ	large enough positive numbers
SR_g	achievable strength ratio for angle g
SRP_i	strength ratio required for part i

Decision Variables

C_{max}	The maximum completion time of any batch on any machine [h]
d_{ijm}	binary variable, equals 1 if part i is assigned to batch j on machine m; 0, otherwise
z_{jm}	binary variable, equals 1 if any part is allocated to batch j on machine m; 0 otherwise
o_{ig}	binary variable, equals 1 if angle g is selected for part i; 0, otherwise (angle decision variable)

a_{ing}	binary variable, equals 1 if part i is allocated in rotation n at angle g; 0, otherwise (rotation decision variable)
b_{ik}	binary variable, equals 1 if part i is located at the left of part k; 0, otherwise
c_{ik}	binary variable, equals 1 if part i is located in front of part k; 0, otherwise
or_i	binary variable, equals 1 if part i is located such that its width is parallel to the length of the machine; 0 if the length of part i is parallel to the length of the machine
(x_i, y_i)	x-axis and y-axis coordinates of the front-left corner of part i [cm]
u_{jm}	maximum height of the parts allocated to batch j on machine m [cm]
p_{jm}	processing time of batch j on machine m [h]
CT_{jm}	completion time of batch j on machine m [h]
δ_i	completion time of part i [h]
T_i	tardiness of part i [h]
T	total tardiness [h]

The bi-objective MILP model here proposed aims to 1) minimize the makespan, C_{max}, which is the maximum completion time of any batch on any machine (Eq. (1)) and 2) minimize the total tardiness, T, defined as the sum of the total tardiness of each part, T_i, as reported in Eq. (2).

$$\text{Min } C_{max} \tag{1}$$

$$\text{Min } \mathrm{T} = \sum_i T_i \tag{2}$$

Eq. (1) and Eq. (2) are the two objective functions of our model while the following equations represent the set of constraints. Each part, *i*, must be allocated exactly to one batch, *j*, and machine, *m*. This is set by considering Eq. (3), where for each part *i* the sum among all batches and machines of binary variables d_{ijm} (1 if part *i* is assigned to batch *j* and machine *m*, 0 otherwise) must be equal to 1.

$$\sum_{j \in J} \sum_{m \in M} d_{ijm} = 1 \; \forall \, i \in I \tag{3}$$

Only a production angle, *g*, can be selected for each part *i* as defined in Eq. (4). Similarly to Eq. (3), a binary variable, o_{ig}, is introduced and it assumes a value of 1 if for the part *i* the production angle *g* is selected or 0 otherwise.

$$\sum_{g \in G} o_{ig} = 1 \; \forall \, i \in I \tag{4}$$

For the nesting problem only one rotation angle, *n*, must be selected for each part, *i*, with a defined production angle, *g*. This is set by introducing the binary variable a_{ing} which assumes a value equal to 1 when rotation *n* is selected for part *i* with production angle *g* and 0 otherwise. The constraint is defined in Eq. (5).

$$\sum_{n\in N}\sum_{g\in G_{in}} a_{ing} = 1\ \forall\ i \in I \qquad (5)$$

Contrarily to the existing works, in our model we aim to include the strength ratio required by the costumers for the printed part. Thus, a constraint that guarantees the respect of the required strength ratio, SRP_i , is necessary. For each part, *i*, the production angle and its related strength ratio, SR_g, must be greater than or equal to the SRP_i as defined by Eq. (6).

$$\sum_{g\in G} SR_g o_{ig} \geq SRP_i\ \forall\ i \in I \qquad (6)$$

Two additional constraints are necessary to guarantee that 1) for each part, *i*, and production angle, *g*, a rotation angle, a_{ing}, is selected and 2) for each batch, *j*, and machine, *m*, at least one part, *i*, must be present. These constraints are defined in Eq. (7) and Eq. (8) by using the big-M value, ψ.

$$\sum_{n\in N} a_{ing} \leq \psi o_{ig}\ \forall\ i \in I, g \in G \qquad (7)$$

$$\sum_{i\in I} d_{ijm} \leq \psi z_{jm}\ \forall\ j \in J, m \in M \qquad (8)$$

For the scheduling problem, and, thus, the definition of the makespan, C_{max}, the processing time of each batch, *j*, on each machine, *m*, must be computed by considering the set-up time of each machine, SET_m, the printing time (which depends on the volumetric printing time VT_m and on the volume of the parts allocated to the batch, including the required support structures), and the powder recoating time (which depends on the recoater time HT_m and on the maximum height of the parts allocated to the batch). For each batch, *j*, and machine, *m*, the processing time p_{jm} is defined in Eq. (9).

$$p_{jm} = SET_m\, z_{jm} + VT_m \sum_{i\in I} d_{ijm}\left(v_i + \sum_{n\in N}\sum_{g\in G} a_{ing} s_{ing}\right) + HT_m \frac{u_{jm}}{lt}\ \forall\ j \in J, m \in M \qquad (9)$$

Then, for each batch, j, in each machine, *m*, the completion time, CT_{jm}, must be computed by considering the previous batch completion time, $CT_{(j-1)m}$ and the processing time of the batch j, p_{jm}. Further, a batch, *j*, cannot start prior to the release time of each part allocated to that batch, r_i. Finally, once the completion time of each batch is defined, the makespan, C_{max}, can be computed by considering the maximum completion time for all batches and machines, CT_{jm}. These constraints are defined in Eq. (10), Eq. (11) and Eq. (12), respectively.

$$CT_{(j-1)m} + p_{jm} \leq CT_{jm}\ \forall\ j \in J\backslash\{1\}, m \in M \qquad (10)$$

$$r_i d_{ijm} + p_{jm} \leq CT_{jm}\ \forall\ i \in I, j \in J, m \in M \qquad (11)$$

$$CT_{jm} \leq C_{max}\ \forall\ j \in J, m \in M \qquad (12)$$

In our model we have two objective functions, the makespan minimization and the total tardiness minimization. To compute the total tardiness, T, we must quantify the tardiness of each part, T_i, by computing the completion

time for each part, δ_i, by considering the batch, j, where is part, i, has been allocated. Finally, the tardiness of the part *i* can be computed by subtracting to the completion time, δ_i, the due date of the part, dd_i.
This corresponds to Eq. (13) and Eq. (14).

$$\delta_i \geq CT_{jm} - \psi\left(1 - d_{ijm}\right) \; \forall \, i \in I, j \in J, m \in M \tag{13}$$

$$T_i \geq \delta_i - dd_i \; \forall \, i \in I \tag{14}$$

Moving on to the nesting problem, it is necessary to include some constraints to guarantee that a part can only be located within the machine's platform. This is defined by adding Eqs. (15) – Eq. (18) in the mathematical model.

$$x_i + \sum_{n \in N} \sum_{g \in G} a_{ing} \, w_{ing} \leq W_m + \psi\left(1 - d_{ijm} + or_i\right) \; \forall \, i \in I, j \in J, m \in M \tag{15}$$

$$x_i + \sum_{n \in N} \sum_{g \in G} a_{ing} \, l_{ing} \leq W_m + \psi\left(2 - d_{ijm} - or_i\right) \; \forall \, i \in I, j \in J, m \in M \tag{16}$$

$$y_i + \sum_{n \in N} \sum_{g \in G} a_{ing} \, l_{ing} \leq L_m + \psi\left(1 - d_{ijm} + or_i\right) \; \forall \, i \in I, j \in J, m \in M \tag{17}$$

$$y_i + \sum_{n \in N} \sum_{g \in G} a_{ing} \, w_{ing} \leq L_m + \psi\left(2 - d_{ijm} - or_i\right) \quad \forall \, i \in I, j \in J, m \in M \tag{18}$$

Further, we must prevent any two parts from overlapping with each other in case they are allocated to the same batch. Thus, the equation set (19)-(23) must be included in the model.

$$\mathrm{x_i} + \sum_{\mathrm{n} \in \mathrm{N}} \sum_{\mathrm{g} \in \mathrm{G}} \mathrm{a_{ing}} \, \mathrm{w_{ing}} \leq \mathrm{x_k} + \psi(1 - \mathrm{b_{ik}} + \mathrm{or_i}) \; \forall \, \mathrm{i}, \mathrm{k} \in \mathrm{I}, \mathrm{i} \neq \mathrm{k} \tag{19}$$

$$x_i + \sum_{n \in N} \sum_{g \in G} a_{ing} \, l_{ing} \leq x_k + \psi(2 - b_{ik} - or_i) \; \forall \, i, k \in I, i \neq k \tag{20}$$

$$y_i + \sum_{n \in N} \sum_{g \in G} a_{ing} \, l_{ing} \leq y_k + \psi(1 - c_{ik} + or_i) \; \forall \, i, k \in I, i \neq k \tag{21}$$

$$\sum_{n \in N} \sum_{g \in G} a_{ing} \, w_{ing} \leq y_k + \psi(2 - c_{ik} - or_i) \; \forall \, i, k \in I, i \neq k \tag{22}$$

$$y_i + b_{ik} + b_{ki} + c_{ik} + c_{ki} \geq d_{ijm} + d_{kjm} - 1 \; \forall \, i, k \in I, i < k, j \in J, m \in M \tag{23}$$

The height of each batch, *j*, must be lower or equal to the maximum height of the selected machine, *m*. Further, the height of each batch, j, is influenced by the height of the parts belonging to it. These constraints are defined by Eq. (24) and Eq. (25).

$$\sum_{n \in N} \sum_{g \in G} a_{ing} \, h_{ing} \leq u_{jm} + \psi\left(1 - d_{ijm}\right) \; \forall \, i \in I, j \in J, m \in M \tag{24}$$

$$u_{jm} \leq H_m \; \forall \, j \in J, m \in M \tag{25}$$

Finally, the domain of each variable must be adequately set as follows:

$$z_{jm} \in \{0,1\} \; \forall \, j \in J, m \in M \tag{26}$$

$$d_{ijm} \in \{0,1\} \; \forall \, i \in I, j \in J, m \in M \tag{27}$$

$$o_{ig} \in \{0,1\} \; \forall \, i \in I, g \in G \tag{28}$$

$$o_i \in \{0,1\} \; \forall \, i \in I \tag{29}$$

$$a_{ing} \in \{0,1\} \; \forall \, i \in I, n \in N, g \in G \tag{30}$$

$$\delta_i, T_i, x_i, y_i \geq 0 \; \forall \, i \in I \tag{31}$$

$$u_{jm}, CT_{jm}, p_{jm} \geq 0 \; \forall \, j \in J, m \in M \tag{32}$$

Eq. (9), which sets the processing time of each batch, is not linear. Hu et al. (2022) proposed a linearization by replacing p_{jm} in Eq. (9) with the following equation (Eq. (33)) and by adding an additional set of constraints which correspond to the linearization (Eq. (34)). Finally, the variable domain is set in Eq. (35).

$$p_{jm} = SET_m z_{jm} + VT_m \sum_{i \in I} \left(v_i d_{ijm} + e_{ijm} \right) + HT_m \frac{u_{jm}}{lt}; \; \forall \, j \in J, m \in M \tag{33}$$

$$e_{ijm} \geq \sum_{n \in N} \sum_{g \in G_{in}} a_{ing} \, s_{ing} - \psi\left(1 - d_{ijm}\right); \; \forall \, i \in I, j \in J, m \in M \tag{34}$$

$$e_{ijm} \geq 0; \; \forall \, i \in I, j \in J, m \in M \tag{35}$$

5. Solution approaches

5.1. ε-constraint algorithm

To solve the mathematical model described in the previous section, we apply the ε-constraint algorithm. By doing so, it is possible to obtain the Pareto front and, consequently, the set of optimal solutions. With the ε-constraint algorithm, the bi-objective problem is reduced to a single-objective one by adding an additional constraint representing one of the objective functions (Haimes et al., 1971). In this specific case, the ε-constraint algorithm can be described in two steps:

- **Step 1**: an upper bound of the total tardiness is fixed equal to $\overline{TT} = UB$, and the MILP problem denoted 3DNS-TT defined with the set of equations: $\sum_i T_i \leq \overline{TT}$; {(1); (3)-(8); (10)-(35)} is solved. 3DNS-TT determines a solution by respecting the fixed maximum value that the total tardiness can assume while minimizing the makespan, C'_{max}.
- **Step 2**: the optimal C'_{max} obtained in Step 1 is fixed as a bound, and the problem denoted as 3DNS-C_{max}, defined by the set of equations: $CT_{jm} \leq C'_{max}$; {(2); (3)-(8); (10); (11); (13)-(35)} is solved. In this way, the non-dominated point with respect to the fixed $\overline{TT}$ can be obtained. The algorithm reduces the upper bound UB on the TT by one unit and goes back to Step 1.

The stopping condition is reached when the bound on the total tardiness cannot be further reduced (i.e., $\sum_i T_i \leq TT_{min}$ with $TT_{min} = 0$). It corresponds to the situation when all parts have no tardiness. The detailed pseudocode of the algorithm is defined in Algorithm 1.

Algorithm 1: ε-constraint algorithm for 3DNS	
1	F=∅; i←0
2	Set $\overline{TT}$←UB
3	**while** ($TT_{min} \leq \overline{TT}$) **do**
4	Set C'_{max} ←Solve 3DNS-TT

    Set $\overline{TT}_i$ ←Solve 3DNS-C_{max}
  S←S∪$\{C'_{max}; \overline{TT}_i\}$
  Decrease the bound on the budget by 1 unit: $\overline{TT} \leftarrow \overline{TT}_i - 1$
  i ←i+1
  end while
  return F   ⇒ return the Pareto front F

5.2. Non-dominated sorting genetic algorithm-II (NSGA-II)

NSGA-II, developed by Deb et al. (2002), is a well-known multi-objective evolutionary algorithm (Yi-Kuei & Cheng-Ta, 2012; Rabbani et al., 2019). The algorithm is quite fast and efficient for multi-objective optimization and has been utilized for a wide range of problems (Rohaninejad et al., 2021). The following subsections provide the main flow of the proposed NSGA-II method with problem-specific solution representation and decoding procedures.

5.2.1. Solution representation

Due to the unique feature of the problem studied in this paper, a paired-gene chromosome representation is utilized in the NSGA-II proposed in this research. The build preference (BF) determined for a part is located just beside that part on the chromosome. Thus, a feasible BF (randomly determined respecting the strength ratio of the part) is inserted into the chromosome for each part, and the length of the chromosome reaches $2 * nbParts$. Specifically, each part can be built in three different angles (g), i.e., 0^0, 45^0, and 90^0 and three different rotations (n), i.e., 1, 2, 3, which yields nine possible combinations. Each of these combinations refers to a BF as shown in Table 2, to simplify the representation of the chromosome. Thus, the BF of each part is also stored on the chromosome to determine the rotation and production angle of the part. For example, rotation 1 for angle 45^0 corresponds to $BF = 6$.

Table 2. The correspondence of the build preference with angle and orientation

Angle	90°			0°			45°		
Rotation	1	2	3	1	2	3	1	2	3
Build Preference BF	0	1	2	3	4	5	6	7	8

When building a new chromosome, this research utilizes a structured procedure with the aim of a good and quick start. For this aim, the geometric means of the release times and due dates ($geo_i = \sqrt{r_i \cdot dd_i}$) are calculated and used during chromosome generation. All parts are listed in the ascending order of geo_i, and the list is separated into s sections. The value of s is determined by the number of parts $\lfloor\sqrt{nbParts}\rfloor$), where $\lfloor X \rfloor$ represents an integer number lower than or equal to X. This aims to keep a balance between the length of the chromosome and the number of parts in each section. Each section is shuffled, and all sections are combined to have a complete sequence of Chr. Now, the length of the chromosome is equal to $nbParts$.

Table 3. Sample data for 10 parts

i	r_i	dd_i	SRP_i	geo_i
1	1	35	55%	5.92
2	0.5	75	88%	6.12
3	0.5	69	71%	5.87
4	1	61	68%	7.81
5	2	108	94%	14.70

6	6	81	50%	22.05
7	5	95	83%	21.79
8	8	71	83%	23.83
9	7	116	57%	28.50
10	10	114	70%	33.76

To give an example, Table 3 represents sample data (including the release dates, due dates and requested strength ratios) for 10 parts. The geo_i value of each part is also calculated using the provided r_i and dd_i, given in the last column. As seen, the minimum and maximum geo_i values are $geo_3 = 5.87$ and $geo_{10} = 33.76$, respectively. To generate a chromosome, the parts are sorted in the ascending order of geo_i, i.e., [3, 1, 2, 4, 5, 7, 6, 8, 9, 10] and partitioned into $s = \lfloor\sqrt{10}\rfloor = 3$ sections: [3, 2, 1], [4, 5, 7], and [6, 8, 9, 10]. Each section is shuffled and combined, which may result in a possible completed sequence of [2, 1, 3, 4, 7, 5, 10, 9, 6, 8]. This is followed by obtaining a paired-gene representation by inserting a BF next to each gene (part). For example, the first gene (part 2) requires a minimum strength ratio of 88%, which can be built with either a 0^0 angle or a 45^0 angle. Thus, a $BF = 7$ for part 2 is randomly determined between [3, 8] and inserted into the chromosome next to part 2. This is followed by the generation of a feasible BF for the subsequent part (i.e., 1) on the chromosome. Namely, part 1 can be fabricated in any of the angles $SRP_1 = 55\%$. Thus, a random $BF = 4$ is generated between [0, 8] and inserted next to 1. This procedure is repeated for all parts, and eventually, a complete chromosome is generated as [2, 7, 1, 4, 3, 4, 4, 4, 7, 6, 5, 8, 10, 7, 9, 4, 6, 7, 8, 5]. Now, the chromosome length reaches to $2 * nbParts$. This procedure allows for the generation of diversified chromosomes satisfying the mechanical properties of the parts as a result of the production angle and rotation. For example, another randomly generated feasible chromosome would be [3, 5, 1, 0, 2, 3, 4, 5, 7, 3, 5, 6, 8, 6, 10, 8, 9, 8, 6, 2].

5.2.2. Decoding procedure

The procedure given in Algorithm 2 is used to calculate the objective values of each chromosome in the population and those generated as a result of genetic operators. Although the procedure is guided to possibly lead to better assignment solutions, randomness is also allowed to avoid being stuck in local optima. The first gene (part number) is assigned to the first batch of an arbitrary (but feasible) machine, considering the BF accompanying it. Note that a part may not be suitable to be assigned to a machine (due to the dimensions) for a certain BF, while it would be suitable for a different BF. This random start allows the diversification of the population for the exploration and exploitation of the search space. The *guided dispatch rule* is utilized as follows for the second and each succeeding part. Available batches are determined (considering the current batches of machines), and the part is assigned to the most favorable one that yields a minimum CT_{jm} and $totTard_{jm}$ value (in the case of assignment). Notice that there might already be assigned parts in a batch (j, m). Therefore, the newly assigned part increases CT_{jm} due to the increase in the volume. It may also increase the release date and the maximum height of the batch. This affects the completion times of the parts already assigned in the batch and so $totTard_{jm}$. If one machine is favorable in terms of CT_{jm} while the other one gives a better $totTard_{jm}$, the part is assigned to one of these machines with an equal chance. Values of the corresponding batch parameters (u_{jm}, v_{jm}, p_{jm}, and CT_{jm}) are updated every time a part is assigned, and the part and its BF are removed from the chromosome. Upon the assignment of all parts on the chromosome, the values of the makespan (f_1) and the total tardiness (f_2) can easily be obtained for the chromosome.

Algorithm 2: Pseudocode of the decoding mechanism

Input: Chr
Reset batch parameters and objective values
Set the current batch index for each AM machine to 1
$ChrTemp \leftarrow Chr$ //create a sample of the chromosome
Get the first gene (part) and the second gene (<u>BF</u>) on the chromosome
Retrieve BF-independent parameters for the part, namely r_i, v_i and dd_i
Retrieve BF-dependent parameters for the part, namely h_{ing}, w_{ing}, l_{ing}, and s_{ing}
<u>Determine available batches</u> (and machines), <u>$AvailBat$, ensuring $nesting()$</u>
Select an arbitrary machine and assign the product to the current batch of the selected machine
Update u_{jm}, v_{jm}, p_{jm}, and CT_{jm} based on the assigned product
Remove the first and the second genes from $ChrTemp$
while $len(ChrTemp) > 0$ //$len(ChrTemp)$ *denotes the length of the* $ChrTemp$
    Get the next gene (part) and its succeeding gene (<u>BF</u>) on the chromosome
    Retrieve BF-dependent and BF-independent parameters for the part
    <u>Determine available batches</u> (and machines), <u>$AvailBat$, ensuring $nesting()$</u>
    //select a machine
    Generate a random number (r) between [0,1]
    if ($r < R2$) and (<u>$AvailBat \neq \emptyset$</u>) //assign to a current batch
        if $len(AvailBat) \neq 1$
            Calculate possible CT_{jm} and $totTard_{jm}$ values for the current batches of each machine in <u>$AvailBat$</u>
            Select the machine that yields the minimum CT_{jm} and $totTard_{jm}$ value
        else
            Select the only machine in $AvailBat$
    else //assign to a newly opened batch
        Select the machine which yields the minimum CT_{jm} when the part to be assigned in the subsequent batch
        Open a new batch on the selected machine and increase the current batch index
    Assign the part to the current batch of the selected machine
    Update u_{jm}, v_{jm}, p_{jm}, and CT_{jm} based on the assigned part
    Remove the assigned part and its BF from $ChrTemp$
    end if
end while
Calculate objective values based on the assignment solution obtained
return f_1, f_2

When determining the available batches, the $nesting()$ function (Andersson & Condello, 2023) is called to ensure that the parts can be placed on the building platform with no overlap. It is obvious that this process is very time consuming, so we utilize some procedures to speed it up. First, this function is skipped if the machine dimensions allow, and the batch is empty (a *true* value is returned without calling the function). Second, if the remaining area of the batch is not large enough to assign the part, then it is obvious that the part cannot be placed with no overlap. In this case, the function is not called; instead, a *false* value is returned.

Given that the method proposed in this research seeks solutions to minimize two objective functions, a *stochastic batch opening policy* is adopted. For this aim, a random number (r) between [0, 1] is determined, and the part is assigned to an existing batch with a predetermined percentage ($R2$). Another situation in which a new batch must be opened is when there is no available batch to which the part can be assigned. In these cases, a batch is opened on the machine yielding the minimum CT_{jm}, and the part is assigned to the newly opened batch.

5.2.3. Genetic operators

Crossover and mutation are applied to generate new individuals from the existing population. A two-point crossover is applied to combine the genetic characteristics of two parents ($parent1$ and $parent2$) sustaining feasibility. Each parent is selected randomly, and two random cut points are determined such that $cp_1, cp_2 = 2 * rand[1, nbParts - 1]$. It is ensured that the parts are not separated from their BF. The head (genes between 0 and $cp1$) and tail (genes between $cp2$ and $nbParts$) are directly taken from $parent1$ to obtain $child1$. The missing genes between $cp1$ and $cp2$ are retrieved as per their sequence on $parent2$. Thus, each part is represented on the chromosome only once, maintaining the permutation-based coding structure. Similarly, $child2$ is obtained by the head and tail parts of $parent2$, and the missing genes are completed based on their sequence on $parent1$.

Mutation is utilized via four operators, *swap*, *insert*, *scramble*, and *random-BF*. These operators are applied with an equal chance. In *swap*, two pairs of the part and BF are determined randomly, and their locations on the chromosomes are swapped with each other. A randomly determined pair of part-BF is relocated to another randomly selected location in the *insert* operator. The *scramble* operator shuffles the sequence of a randomly determined section of the chromosome, binding the BFs to the parts. The last operator, *random-BF*, alternates the BF values of two randomly determined parts, ensuring feasibility.

5.2.4. Forming the new generation

Upon obtaining the children from the genetic operators, their fitness values are calculated using the decoding procedure given in Section 5.2.2. This is followed by removing the clones in children, which have the same fitness values with each other or any chromosome in the population, to avoid being stuck in local optima. Afterward, a new set of solutions are obtained, combining the existing population and newly obtained children, and NSGA-II's non-dominated sorting and crowding distance operators are utilized. This is intended to determine the new population for the next iteration of the algorithm. The non-dominated sorting operator determines the fronts, and subsequently the crowding distance operator evaluates each solution in the same front. Solutions having the maximum distance are more favorable to other solutions in the same non-dominated set of solutions. The next generation is formed starting from the first front (i.e., Pareto front), and $popSize$ chromosomes are included in the population. If $popSize$ is exceeded when the next front is added, the new population is completed by taking the best solutions according to the crowded-distance value from the next front. A sample evolution of the population through generations for one test problem (i.e., P14, to be presented in the following section) is also given in Figure 2.

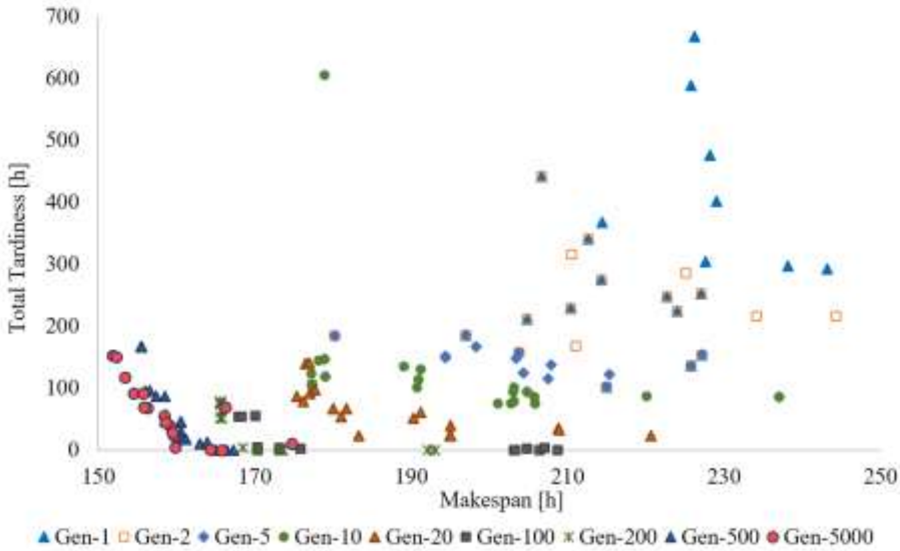

Figure 2. Evolution of a population for a variant of NSGAII (i.e., NSGAII-GO, to be explained in Section 6) when solving P14

5.2.5. General outline

The general flow of the proposed NSGA-II is presented in Algorithm 3. As can be seen, the algorithm starts with initializing parameters, and the initial population is obtained by building randomly generated feasible chromosomes (Chr), as explained in Section 5.2.1. Each chromosome is sent to decoding (see Algorithm 2), and the fitness values (fit_{Chr}) are returned. Eventually, Chr and fit_{Chr} are added to pop and $popFitness$, respectively. This cycle is repeated until an initial population with $popSize$ chromosomes is created.

Algorithm 3: Pseudocode of the proposed NSGA-II

Initialize Parameters
// generate initial population
$pop \leftarrow \{\}, popFitness \leftarrow \{\}$
while ($len(pop) < popSize$) //$len(pop)$ *denotes the length of the population*
    List all parts in ascending order based on the geometric means of the release times and due dates
    Divide parts into s sections keeping their orders, where $s = \lfloor \sqrt{nbParts} \rfloor$
    Shuffle each section to get a partial chromosome Chr'_p, where $p = \{1,2, \dots, s\}$
    Combine the partial sequences to get a complete chromosome $Chr = [Chr'_1 \cup Chr'_2 \cup \dots Chr'_s]$
    foreach part in chromosome Chr
        Insert a feasible randomly generated build preference (BF) beside each part on the chromosome (so the length of the chromosome is $2 * nbParts$)
    end foreach
    $fit_{Chr} \leftarrow decoding(Chr)$
    Add Chr to pop
    Add fit_{Chr} to $popFitness$
end while
$NS\_Operators(pop, popFitness)$
$iter \leftarrow 1$
while ($iter < maxNbIter$)
    $resetIterParam()$

```
        crossover() and mutation() to get children and childFitness
        removeClones() //delete duplications to preserve diversity
        Form a new generation among existing pop and children using NS_Operators
        if the minimum value of any of the objectives is improved
            iter ← 1
        else
            iter ← iter + 1
        end if
    end while
    return the Pareto front
```

All the chromosomes in the initial population are sent to $NS\_Operators$, namely $nondom\_sort()$ and $crowd\_dist()$ to determine the Pareto fronts (given in Section 5.2.4). One can refer to Deb et al. (2002) for the pseudo-codes of the non-dominated sorting and crowding distance operators, which are not provided here due to page limitations. Following the $NS\_Operators$, iterations start to evolve the population through crossover and mutation. The algorithm applies a two-point crossover and four different mutation strategies (with equal chance), as explained in Section 5.2.3. To maintain diversity and avoid being stuck in local optima, a specialized function named $removeClones()$ is applied after children are obtained and their fitness values are calculated. This function eliminates children having the same fitness values as any of the chromosomes in the population. This is needed because the assignment solutions of the chromosomes could be the same, even if their sequences and/or build preferences differ. At this stage, new fronts are determined considering all the chromosomes in the population and children, and the new population is determined following the procedure presented in Section 5.2.4.

Computational experiments are carried out to assess the performance of the algorithms presented here. Additionally, some numerical instances are evaluated to derive managerial insights, as reported in the next section.

6. Computational experiments and analysis

We conducted several computational experiments to evaluate and compare the performances of the proposed algorithm. Further, we focused on an optimally solved instance to derive some managerial insights. We elaborate better on this below. The MILP model is solved with ILOG CPLEX 22.11, and the computing time for each run and instance is limited to 1,800 CPU seconds. The NSGA-II algorithm is coded in Python 3.11 and implemented on a PC equipped with Intel i7 2.8 GHz & 20 GB RAM.

6.1. Experimental design

We generated a total of 56 test instances in different scales, i.e., small, medium, and large. Data on the parts (volume, required strength ratio, due dates, width, length, height, and support volume) were provided by various AM companies. They also provided the machines' features, which were integrated with those reported in Kucukkoc (2019). The details of both parts and machines are available as supplementary material (see SM1). Notably, four different versions of the proposed NSGA-II were constructed to observe the influences of the proposed dispatch rule and batch opening policy on its performance:

- NSGAII-RF: NSGAII algorithm with the *random dispatch rule* but *without the stochastic batch opening policy*
- NSGAII-RO: NSGAII algorithm with the *random dispatch rule* and the *stochastic batch opening policy*
- NSGAII-GF: NSGAII algorithm with the *guided dispatch rule* but *without the stochastic batch opening policy*
- NSGAII-GO: NSGAII algorithm with the *guided dispatch rule* and the *stochastic batch opening policy*

The random dispatch rule allocates parts to any available batch and machine randomly. However, as explained in Section 5.2.2, the guided dispatch rule allocates parts to the best available batch that yields the minimum objectives (i.e., the makespan and the total tardiness of all parts already assigned). The other mechanism proposed in this research, i.e., the stochastic batch opening policy, opens new batches at a certain probability to minimize the total tardiness as well as the makespan. Otherwise, filling all batches to the maximum capacity might minimize the makespan but contradicts the total tardiness objective. The following section reports the results of the proposed model and meta-heuristics.

6.2. Test results

Each problem was solved multi-objectively five times by running each meta-heuristic algorithm, and the results are presented in Tables 4 and 5. The minimum of the first and second objectives in the Pareto front (i.e., the makespan, f_1, and the total tardiness, f_2) are reported in the columns '*MinF1*' (for the makespan) and '*MinF2*' (for the total tardiness) for each of the four algorithms. Then, we also compare the algorithm in terms of the hypervolume of the Pareto front, which is a well-known metric used in multi-objective optimization (Beume et al., 2007; Audet et al., 2020, Rodríguez-Ballesteros et al. 2024). Specifically, we normalized the hypervolume of each algorithm (by dividing it by the maximum hypervolume of all algorithms for the same problem) and reported it in the column '*MeanHV_N*'. The average CPU time consumed while solving the problems is also provided in the column '*CPU_Av*'. The performances of the proposed meta-heuristics in terms of the performance metrics investigated are also plotted in Figure 3. Notably, since solving all the problems using the ε-constraint algorithm to get the Pareto front takes too much time, small- and medium-size problems were solved 'single-objectively' with the proposed mathematical model (given in Section 4) over CPLEX, and the results are provided in Table 4. The objective values are presented in the columns 'F1S' and 'F2S', together with their gaps and CPU times in the sub-sequencing columns. It is worth noting that F1S [F2S] was obtained when the sole objective of the model was to minimize the makespan [total tardiness]. Therefore, it is important to notice that these results are reported just to give an overall idea regarding the optimal values of the individual objectives when evaluating the performances of the meta-heuristics. They should not be directly compared with MinF1 and MinF2 obtained by meta-heuristics, where both objectives are optimized simultaneously.

When the results in Tables 4 and 5 are examined, it can be seen that the proposed meta-heuristics show relatively similar performance in terms of the minimum objective values in small-size problems. However, when the problem size grows, NSGAII-GF and NSGAII-GO show better performance in comparison with NSGAII-RF and NSGAII-RO. This especially becomes more apparent after P22 for MinF1, and after P26 for MinF2. As per the normalized mean hypervolumes (MeanHV_N), NSGAII-GF and NSGAII-GO clearly outperform NSGAII-RF and NSGAII-RO. Finally, in terms of the average CPU time consumed, NSGAII-GF is generally better than NSGAII-GO, NSGAII-RF and NSGAII-RO for small-size problems. However, when the problem size grows, none of the algorithms stands out above the others. This could be a result of fast convergence thanks to the guided dispatch rule utilized by NSGAII-GF. However, the opportunity of opening new batches at a certain probability leads to relatively increased computation times for NSGAII-GO.

To further elaborate on the performance comparison of the four meta-heuristics, Table 6 presents their pairwise comparisons in terms of MinF1, MinF2, and MeanHV_N. The number given in each cell in the table shows how many problems the algorithm in the relevant row finds better results than the algorithm in the corresponding column. For example, NSGAII-RF finds better MinF1 values than NSGAII-RO for 28 test problems (while it is outperformed by NSGAII-RO for 25 problems). As can be seen from Table 6a, NSGAII-GO leads to a better MinF1 value than NSGAII-RF, NSGAII-RO, and NSGAII-GF for 42, 46, and 28 problems, respectively. Nonetheless, NSGAII-GF outperformed NSGAII-RF, NSGAII-RO, and NSGAII-GO for 41, 43, and 22 problems. Though the statistical test results (given in Appendices) indicate that the difference between the performances of the algorithms are not statistically significant ($p > 0.05$), this finding indicates that NSGAII-GO is the best performer, and NSGAII-GF is the second-best performer in terms of MinF1.

Table 4. Computational test results for small- and medium-size problems

Prob. ID	Single Objective Results						Multi Objective Results															
	CPLEX						NSGAII-RF				NSGAII-RO				NSGAII-GF				NSGAII-GO			
	F1S	GapF1	CPU	F2S	GapF2	CPU	MinF1	MinF2	MeanHV_N	CPU_Av	MinF1	MinF2	MeanHV_N	CPU_Av	MinF1	MinF2	MeanHV_N	CPU_Av	MinF1	MinF2	MeanHV_N	CPU_Av
P1	*109.07*	0.01%	47.53	*9.42*	0%	3.41	**109.82**	18.06	0.9805	38.52	**109.82**	18.06	0.9768	44.75	**109.82**	16.07	0.9719	60.85	**109.82**	**9.61**	**1.0000**	37.71
P2	*115.18*	0.01%	194.61	*23.47*	0%	52.70	119.45	**25.06**	0.9756	41.30	119.20	**25.06**	0.9939	45.50	**118.86**	26.49	0.9293	40.08	**118.86**	**25.06**	**1.0000**	69.90
P3	*109.23*	0.01%	485.15	*9.42*	0%	40.87	**109.82**	18.06	0.9363	40.43	**109.82**	18.06	0.9623	40.99	110.59	18.06	0.9696	37.29	**109.82**	**9.61**	**1.0000**	36.30
P4	*109.08*	0.15%	1800.00	*9.42*	0%	158.84	109.85	18.06	0.9785	70.80	**109.82**	18.06	0.9822	53.42	**109.82**	18.06	0.9668	48.51	**109.82**	**9.88**	**1.0000**	59.93
P5	*116.04*	1.80%	1800.00	*23.47*	0.01%	238.73	**118.86**	25.14	0.9953	57.47	**118.86**	25.14	**1.0000**	63.36	**118.86**	25.34	0.9786	70.90	**118.86**	**25.06**	0.9946	48.85
P6	101.93	1.82%	1800.00	0.53	0.00%	113.71	102.47	3.72	0.9911	101.08	102.47	3.72	0.9887	79.70	101.66	3.28	0.9944	55.87	**101.61**	**0.53**	**1.0000**	96.10
P7	66.69	10.99%	1800.00	*0.00*	0.00%	0.65	66.87	**0.00**	0.9974	53.70	67.65	**0.00**	0.9981	50.31	67.61	**0.00**	0.9931	67.62	**66.41**	**0.00**	**1.0000**	70.88
P8	*130.74*	13.63%	1800.00	18.79	100%	1800.00	132.95	24.68	0.9799	65.33	133.40	24.76	0.9757	53.06	132.25	24.68	0.9822	58.84	**131.86**	**16.62**	**1.0000**	106.43
P9	130.85	14.29%	1800.00	*16.68*	100%	1800.00	133.77	25.29	0.9587	70.71	**129.12**	25.27	0.9818	81.98	133.51	25.06	0.9579	81.04	129.30	**16.70**	**1.0000**	127.48
P10	*110.63*	11.74%	1800.00	0.00	0%	11.07	112.59	2.71	0.9546	102.38	112.04	1.94	0.9665	97.35	**111.64**	5.06	0.9684	71.94	112.75	**0.00**	**1.0000**	67.70
P11	117.69	10.87%	1800.00	*2.96*	100%	1800.00	121.06	8.78	0.9222	72.81	124.71	8.78	0.9147	91.44	117.89	8.78	**1.0000**	67.49	**116.27**	**7.47**	0.9986	90.28
P12	*169.72*	15.44%	1800.00	20.72	100%	1800.00	**173.71**	0.06	0.9974	91.80	180.51	**0.00**	0.9710	99.45	175.02	**0.00**	0.9620	87.13	175.83	**0.00**	**1.0000**	104.03
P13	*186.16*	16.77%	1800.00	40.12	100%	1800.00	191.77	**16.80**	0.9875	64.59	191.77	**16.80**	**1.0000**	77.83	**191.69**	**16.80**	0.9888	59.66	**191.69**	**16.80**	0.9883	57.38
P14	148.35	15.60%	1800.00	0.00	0%	22.88	155.10	**0.00**	0.9216	85.08	154.01	**0.00**	0.9101	130.40	148.20	**0.00**	0.9344	82.18	**145.99**	**0.00**	**1.0000**	142.88
P15	*98.61*	16.48%	1800.00	0.00	0%	2.31	100.70	**0.00**	**1.0000**	96.44	100.73	**0.00**	0.9990	74.41	**99.88**	**0.00**	0.9969	84.06	**99.88**	**0.00**	0.9872	104.54
P16	*180.84*	19.20%	1800.00	19.75	100%	1800.00	**182.93**	0.46	0.9976	637.40	186.46	1.45	0.9801	231.75	184.57	**0.00**	0.9973	193.14	184.22	**0.00**	**1.0000**	310.45
P17	*193.74*	19.60%	1800.00	47.85	100%	1800.00	202.05	25.53	**1.0000**	104.19	**201.63**	25.43	0.9934	283.61	202.70	25.61	0.9907	160.17	202.22	**22.80**	0.9985	278.62
P18	158.15	19.08%	1800.00	0.00	0%	47.55	161.39	**0.00**	0.8920	155.98	162.98	**0.00**	0.8815	171.83	**151.97**	**0.00**	0.9752	186.93	152.07	**0.00**	**1.0000**	97.00
P19	*168.20*	20.26%	1800.00	2.96	100%	1800.00	169.96	**0.00**	0.9645	230.42	172.55	**0.00**	0.9433	185.83	**168.39**	**0.00**	**1.0000**	253.48	169.92	**0.00**	0.9994	95.35
P20	*204.87*	22.10%	1800.00	37.74	100%	1800.00	**208.45**	37.90	0.9628	147.96	208.91	23.48	0.9905	292.01	208.95	28.44	0.9721	249.74	208.82	**19.29**	**1.0000**	258.14
P21	*218.14*	21.74%	1800.00	63.65	100%	1800.00	**221.31**	49.94	0.9873	233.47	222.17	47.16	0.9806	250.96	222.21	**46.96**	0.9790	192.44	221.43	48.77	**1.0000**	284.68
P22	*186.51*	20.10%	1800.00	6.36	100%	1800.00	195.48	**0.00**	0.9668	331.50	194.22	1.04	0.9740	460.37	194.60	**0.00**	0.9814	176.73	**194.19**	**0.00**	**1.0000**	335.78
P23	119.47	30.32%	1800.00	0.00	0%	19.49	119.44	**0.00**	0.9783	384.76	119.79	**0.00**	0.9749	213.85	119.85	**0.00**	0.9695	224.10	**115.92**	**0.00**	**1.0000**	231.55
P24	191.53	26.64%	1800.00	0.00	0%	327.15	189.55	**0.00**	0.8823	230.17	190.53	**0.00**	0.8598	246.31	181.83	**0.00**	0.9621	225.34	**173.29**	**0.00**	**1.0000**	417.07
P25	124.65	33.53%	1800.00	0.00	0%	66.39	**123.91**	**0.00**	**1.0000**	171.62	125.66	**0.00**	0.9928	205.06	125.12	**0.00**	0.9995	317.57	126.32	**0.00**	0.9972	415.17
P26	*132.96*	31.43%	1800.00	0.00	0%	384.44	201.24	1.17	0.9600	312.43	200.01	**0.00**	0.9983	353.13	**198.72**	3.78	0.9965	394.38	201.38	**0.00**	**1.0000**	171.35
P27	213.81	25.87%	1800.00	27.53	100%	1800.00	212.23	21.69	0.9850	286.06	213.86	25.96	0.9364	260.59	212.20	**21.62**	0.9742	479.23	**204.58**	24.57	**1.0000**	500.48
P28	233.68	27.79%	1800.00	37.34	100%	1801.14	233.59	32.74	0.8783	501.06	224.83	27.73	0.9659	473.39	**214.81**	19.93	0.9625	383.34	217.46	**11.97**	**1.0000**	540.05
P29	145.65	36.58%	1800.00	0.00	0%	666.24	140.99	**0.00**	**1.0000**	430.14	140.94	**0.00**	0.9944	344.87	141.21	**0.00**	0.9880	376.41	**140.44**	**0.00**	0.9759	402.05
P30	218.54	26.43%	1800.00	17.14	100%	1800.00	214.71	14.35	0.8597	516.70	215.45	7.32	0.8218	639.18	**200.23**	**0.00**	0.9790	494.31	200.28	**0.00**	**1.0000**	672.40
P31	248.44	27.93%	1800.00	70.95	100%	1800.00	244.68	**53.46**	0.9963	629.09	245.37	65.09	0.9802	605.22	238.42	60.33	0.9861	356.66	**237.42**	58.54	**1.0000**	382.20
P32	147.16	33.36%	1800.00	0.00	0%	854.28	145.87	**0.00**	0.9942	403.42	145.62	**0.00**	0.9853	303.38	**141.92**	**0.00**	**1.0000**	362.56	142.43	**0.00**	0.9878	462.56

** Best values by variants of NSGA-II are in bold. Single objective results better than multi-objective are in italic.*

Table 5. Computational test results for large-size problems

Prob. ID	Multi Objective Results NSGAII-RF				NSGAII-RO				NSGAII-GF				NSGAII-GO			
	MinF1	MinF2	MeanHV_N	CPU_Av	MinF1	MinF2	MeanHV_N	CPU_Av	MinF1	MinF2	MeanHV_N	CPU_Av	MinF1	MinF2	MeanHV_N	CPU_Av
P33	418.15	266.55	0.9189	1250.36	415.88	314.40	0.9536	1235.04	412.75	256.26	0.9820	594.54	**411.45**	**230.28**	**1.0000**	1505.71
P34	255.51	**0.00**	0.9819	677.17	258.54	**0.00**	0.9750	648.23	**245.90**	**0.00**	**1.0000**	307.72	251.45	**0.00**	0.9977	805.41
P35	391.06	174.76	0.8477	936.92	389.71	172.13	0.8700	1111.34	370.68	**70.42**	0.9811	386.81	**364.10**	96.69	**1.0000**	879.24
P36	241.10	**0.00**	0.9953	737.34	240.62	**0.00**	1.0000	680.26	241.10	**0.00**	0.9956	378.62	**234.28**	**0.00**	0.9996	580.75
P37	**245.62**	**0.00**	0.9975	618.10	246.20	**0.00**	0.9902	619.88	249.54	**0.00**	0.9901	273.28	245.71	**0.00**	**1.0000**	1412.42
P38	366.03	65.85	0.8906	1013.56	370.30	109.21	0.8892	613.95	346.55	23.46	0.9629	157.67	**345.60**	**21.33**	**1.0000**	489.10
P39	293.61	**0.00**	0.9963	943.39	293.46	**0.00**	0.9898	853.12	289.06	**0.00**	0.9952	559.23	**283.50**	**0.00**	**1.0000**	520.90
P40	305.57	**0.00**	0.9929	1019.33	301.39	**0.00**	0.9954	1395.45	299.93	**0.00**	**1.0000**	871.65	**295.66**	**0.00**	0.9988	645.51
P41	456.91	499.07	0.8507	1304.99	454.08	577.98	0.8542	2014.66	**421.63**	**357.22**	**1.0000**	1291.02	430.90	467.90	0.8879	1300.12
P42	282.31	**0.00**	0.9844	1393.39	281.15	**0.00**	0.9888	1085.63	277.07	**0.00**	0.9866	2569.03	**268.88**	**0.00**	**1.0000**	1871.32
P43	285.81	**0.00**	0.9893	1011.43	287.26	**0.00**	0.9869	708.35	**273.47**	**0.00**	**1.0000**	1422.68	279.12	**0.00**	0.9901	1150.95
P44	404.99	299.19	0.8970	1263.33	409.05	228.50	0.8821	821.51	**380.09**	**152.37**	**1.0000**	1665.83	386.89	181.95	0.9255	2080.55
P45	312.07	**0.00**	0.9830	1207.87	312.45	**0.00**	0.9865	1178.44	**302.35**	**0.00**	**1.0000**	1193.74	307.70	**0.00**	0.9903	1420.99
P46	320.89	**0.00**	0.9985	1494.89	321.29	**0.00**	0.9984	800.28	**317.25**	**0.00**	**1.0000**	1379.52	321.23	**0.00**	0.9942	1300.57
P47	332.78	**0.00**	0.9934	1207.43	332.03	**0.00**	0.9964	1102.64	**325.16**	**0.00**	**1.0000**	1858.63	338.51	**0.00**	0.9825	923.46
P48	302.90	**0.00**	0.9940	1490.48	303.21	**0.00**	0.9936	1054.68	300.68	**0.00**	0.9897	1381.02	**296.38**	**0.00**	1.0000	1847.34
P49	305.26	**0.00**	0.9963	1285.97	305.35	**0.00**	0.9979	1355.41	**293.66**	1.86	**1.0000**	1043.57	299.42	**0.00**	0.9983	977.89
P50	434.17	530.07	0.8740	1614.49	429.01	349.70	0.9186	1899.11	**404.25**	**280.31**	**1.0000**	2110.08	407.73	353.85	0.9387	3697.14
P51	421.37	16.38	0.9744	2250.34	416.14	22.97	0.9841	2539.44	**402.50**	**0.00**	**1.0000**	2949.43	413.25	34.55	0.9619	3382.71
P52	415.18	48.53	0.9849	3852.14	417.73	**0.00**	0.9952	4303.36	**407.56**	**0.00**	0.9985	1296.49	412.57	22.18	1.0000	2116.24
P53	431.84	91.43	0.9525	1855.53	431.15	23.26	0.9647	4046.07	**416.01**	**9.31**	**1.0000**	2979.92	418.05	46.86	0.9643	3055.16
P54	405.77	18.38	0.9677	1679.12	404.54	8.37	0.9729	2943.57	394.68	**0.00**	**1.0000**	5277.39	**392.53**	**0.00**	0.9923	3620.01
P55	386.31	**0.00**	0.9985	3078.31	393.83	23.91	0.9731	5084.62	386.55	**0.00**	**1.0000**	3935.13	**379.30**	**0.00**	0.9849	2676.09
P56	393.48	4.05	0.9837	1707.73	400.44	**0.00**	0.9878	3685.15	**380.10**	**0.00**	**1.0000**	2712.26	386.25	12.66	0.9814	3940.57

* *Best values by variants of NSGA-II are in bold*

a)
b)
c)

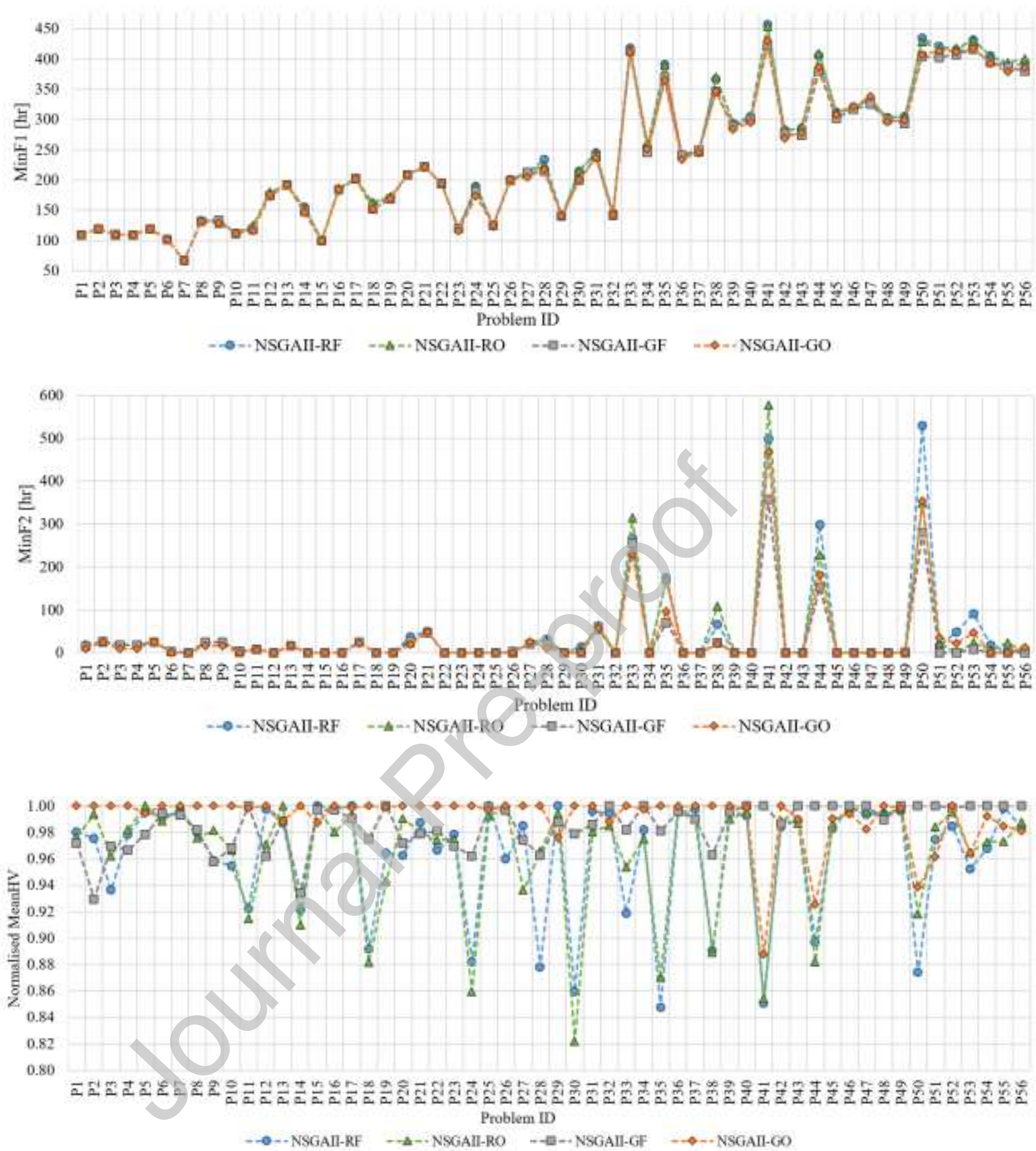

Figure 3. Performance comparisons of the four meta-heuristics in terms of (a) MinF1, (b) MinF2, and (c) Normalized MeanHV

With regards to MinF2 and MeanHV_N, NSGAII-GO finds better solutions than NSGAII-GF for 18 and 34 problems, respectively. On the contrary, NSGAII-GF performs better than NSGAII-GO for 10 and 22 problems with respect to MinF2 and MeanHV_N. Therefore, Tables 6b and 6c indicate that NSGAII-GO shows superiority over NSGAII-GF in a greater number of problems. Furthermore, it was statistically proven that NSGAII-GO is the best performer, and NSGAII-GF is the second-best performer among the four algorithms in terms of MinF2 and MeanHV_N objectives (see Supplementary Material, SM3).

Table 6. Pairwise comparison of the four meta-heuristics in terms of (a) MinF1, (b) MinF2, and (c) MeanHV_N

a)	NSGAII-RF	NSGAII-RO	NSGAII-GF	NSGAII-GO
NSGAII-RF	-	28	12	11
NSGAII-RO	25	-	10	6
NSGAII-GF	41	43	-	22
NSGAII-GO	**42**	**46**	**28**	-

b)	NSGAII-RF	NSGAII-RO	NSGAII-GF	NSGAII-GO
NSGAII-RF	-	10	7	4
NSGAII-RO	17	-	7	6
NSGAII-GF	21	21	-	10
NSGAII-GO	**26**	**24**	**18**	-

c)	NSGAII-RF	NSGAII-RO	NSGAII-GF	NSGAII-GO
NSGAII-RF	-	29	19	11
NSGAII-RO	27	-	20	10
NSGAII-GF	37	36	-	22
NSGAII-GO	**45**	**46**	**34**	-

Despite NSGAII-GF being clearly better than NSGAII-GO for certain specific problems (e.g., P50-P53) in terms of all performance metrics (MinF1, MinF2, and MeanHV_N), the overall evaluation given in Table 6 emphasizes that NSGAII-GO is preferable to NSGAII-GF for MinF1, MinF2 and MeanHV_N. This finding can result in such a consequence that the new stochastic batch opening policy utilized in NSGAII-GO can yield increases in the makespan. However, it could minimize the total tardiness and maximize the hypervolume of the Pareto front. Another important outcome of the summary provided in Table 6 is that the guided dispatch rule provides a clear advantage to NSGAII-GF and NSGAII-GO in terms of minimizing the objective values during decoding.

6.3. Numerical analysis and managerial implications

According to Yu et al. (2022), *"There exists a compromise between the makespan and total tardiness objectives, which originates from the dilemma of 'packing more parts to benefit from the common machine setup/recoating time' or 'packing less parts to maintain the flexibility for handling distributed due dates.'"* In light of this, we are interested in identifying how the makespan and total tardiness minimization influence each other. Indeed, considering only minimization of the total tardiness may yield a solution with a higher makespan and lower machine utilization due to frequent setups, whereas makespan minimization may lead to solutions with higher tardiness for some parts, with a negative impact on the service level and customer satisfaction. To elaborate on this and identify a preferable solution, we apply the efficiency frontier approach, as in Altekin & Bukchin (2022). In particular, we investigate how the number of parts, number of machines, and their features influence the shape of the super Pareto front. Further, we investigate the impact of the strength requirements on the super Pareto front. More specifically, the number of parts, number of machines, and strength requirements were derived from the computational experiments described above and are reported in Table 7, while the machine features are reported in Table 8. As described above, the main goal of this analysis is to derive managerial insights concerning the number of machines, machine features, and strength requirements. To do so, we analyze

12 instances clustered in three groups with a variable number of parts (i.e., 14, 20, 30). For each group, four different combinations of machines are included. For all problems, heterogenous machines sets are considered. Notably, in each group, the same parts are considered. Please note that due to the variability in width, length, and height according to the orientations, we provide only the average volume of the set of parts. Moving on to machines features (Table 8), these are characterized by the same value of the recoated time (it is worth mentioning that this is the time required to spread only one powder layer in the printing area, and that many layers need to be spread, with their number depending on the highest part of the job) but different values of the setup time, volumetric printing time, and height, width, and length of the building chamber, which vary significantly from one machine to another. Particularly, M1 and M4 are larger and higher that M2 and M3 thus we expect to achieve lower makespan and total tardiness values when they are selected. Finally, all parts can be printed in each machine to avoid unfeasible solutions.

Table 7. Instances' features

Group	Problem ID	# parts	Mean Volume [cm^3]	Mean SRP	Mean Release date [h]	Mean Due date [h]	# machine	Machine Type
1	P8	14	7226.814	0.671	7.714	99.642	2	M1, M2
	P9	14	7226.814	0.671	7.714	99.642	2	M1, M4
	P10	14	7226.814	0.671	7.714	99.642	3	M1, M2, M2
	P11	14	7226.814	0.671	7.714	99.642	3	M1, M2, M3
2	P16	20	6759.665	0.646	19.6	144.9	2	M1, M2
	P17	20	6759.665	0.646	19.6	144.9	2	M1, M3
	P18	20	6759.665	0.646	19.6	144.9	3	M1, M2, M2
	P19	20	6759.665	0.646	19.6	144.9	3	M1, M2, M3
3	P29	30	6347.733	0.645	30.533	171.7	3	M1, M1, M2
	P30	30	6347.733	0.645	30.533	171.7	3	M1, M2, M2
	P31	30	6347.733	0.645	30.533	171.7	3	M1, M3, M3
	P32	30	6347.733	0.645	30.533	171.7	3	M1, M1, M3

Table 8. Machines' features

Machine	Setup time [h]	Volumetric printing time [h/cm3]	Recoater time [h/layer]	Height [cm]	Width [cm]	Length [cm]
M1	2	0.001	0.0024	60	60	60
M2	1.5	0.00585	0.0024	36.5	28	50
M3	1.2	0.0104	0.0024	36.5	28	28
M4	2.5	0.00585	0.0024	85	28	50

Aiming to compare points belonging to the same super Pareto front and super Pareto fronts belonging to the same group, we investigated the following key performance indicators (KPIs):

- Ideal solution: refers to the case with the minimum value of the makespan and the total tardiness. Only in the case that the super Pareto front has a single point the ideal point is also the actual solution.
- Extreme points: the two points that refer to the solution with the minimum makespan (and maximum value of the total tardiness) and the minimum total tardiness (and maximum value of the makespan).
- The number of points belonging to the super Pareto front.

- The makespan trade-off refers to the relative difference between the maximum and minimum makespan; notably, the maximum makespan is achieved when the total tardiness is minimized.
- The total tardiness trade-off refers to the relative difference between the maximum and minimum total tardiness; notably, the maximum total tardiness is achieved when the makespan is minimized.
- Distance from the ideal point. It is defined as the relative Euclidean distance of a point belonging to the super Pareto front from the ideal point.
- Preferable solution: the solution with a minimal distance.

For a visual representation of the KPIs, the reader can refer to Altekin & Bukchin (2022).

Figure 4 reports the super Pareto front for all cases here analyzed and clustered into the three groups. As can be seen, the number of machines and features (width, length, height of the building chamber) strongly influence both the makespan and the total tardiness. With regard to the number of machines, as could be expected, the higher the number of machines, the lower the values are of the makespan and the total tardiness. Indeed, focusing, for example, on Group 1 (but the same can be said for Group 2), P8 and P9, where two machines are used, are characterized by higher values of the makespan and the total tardiness than P10 and P11, where three machines are used. It is interesting to observe that the difference between adopting two or three machines is not negligible, with the makespan and the total tardiness being on average 6.09% and up to 78% lower, respectively. Dealing then with the impact of the machine features, we can see from the results that bigger machines should be preferred. Indeed, as can be noted, for example, for both Groups 1 and 2, P11 and P19 lead to the solution with lower total tardiness, while the makespan is slightly higher. This is due to the different values of the setup time and volumetric printing time.

Finally, considering Group 3, we can note a significant difference between P29 and P32, compared to P30 and P31 (cf. Table 9). This is due to the different sizes of the building chambers of different machines: in P30 and P31, smaller machines are mainly used, while in P29 and P32, bigger machines are used. Consequently, P29 and P32 are characterized by lower makespan and total tardiness values since the use of bigger machines allow more parts to be printed on the same job. Hence, bigger machines should be preferred. Moving to the number of non-dominated points of each problem, we cannot identify a trend in terms of how changes in the machine features or number of parts impact the number of non-dominated points. There is only one problem (P32) having, six points, which can be considered as a low number of non-dominated points. In fact, the higher the number of non-dominated points, the higher the complexity is in identifying preferable solutions. Finally, considering all problems investigated here, we can note that the total tardiness is much more influenced than the makespan. Thus, the solution with the minimum makespan is never the preferable solution, as it leads to a very high value of total tardiness, with negative consequence on service level. On the other hand, the preferable solution among all points is always close to the extreme point, which refers to the minimum total tardiness. This is in line with the makespan and the total tardiness trade-offs reported in Table 9.

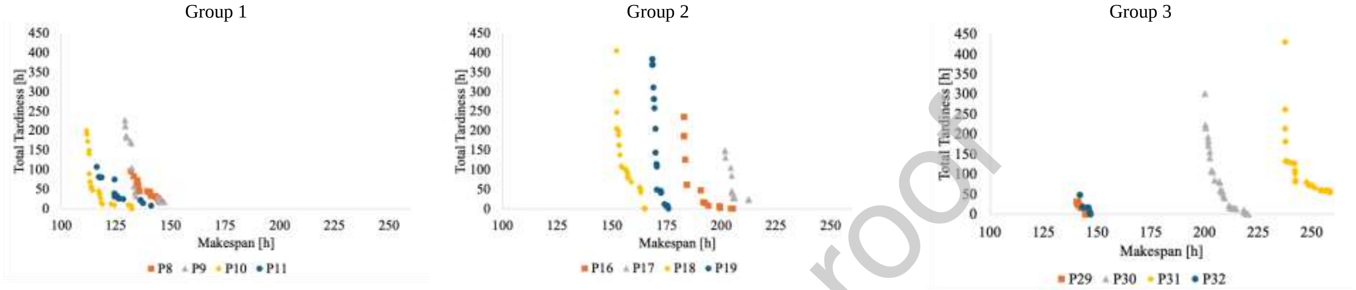

Figure 4. Super Pareto front for the different groups

Table 9. Super Pareto front analysis

Group	# Parts	Problem ID	# Machines	# Point super Pareto front	Makespan trade-off	Total tardiness trade-off	Ideal solution		Preferable solution		
							Makespan	Total tardiness	Distance	Makespan	Total tardiness
1	14	P8	2	21	10.35%	484.27%	131.864	16.618	0.104	145.512	16.618
		P9		21	14.02%	1260.12%	129.118	16.699	0.132	145.502	17.319
		P10	3	21	18.48%	N/D	111.636	0.000	0.185	132.268	0.000
		P11		18	21.29%	1333.46%	116.265	7.471	0.213	141.013	7.471
2	20	P16	2	16	12.19%	N/D	182.927	0.000	0.122	205.223	0.000
		P17		25	5.39%	551.61%	201.628	22.799	0.054	212.494	22.799
		P18	3	28	8.79%	N/D	151.967	0.000	0.088	165.323	0.000
		P19		23	4.39%	N/D	168.393	0.000	0.044	175.793	0.000
3	30	P29	3	14	2.92%	N/D	140.437	0.000	0.029	144.531	0.000
		P30		37	10.06%	N/D	200.228	0.000	0.101	220.375	0.000
		P31		37	8.87%	704.22%	237.417	53.462	0.089	256.529	53.219
		P32		6	3.50%	N/D	141.920	0.000	0.035	146.882	0.000

Further, we want to elaborate on the considerations regarding the strength requirements introduced here for the first time in the literature. Whenever parts are required to withhold loads, AM manufacturers could be tempted to produce them with the maximum achievable strength (i.e., $SR = 1$) to be on the safe side. This would correspond to printing them with a production angle equal to 45°. However, this approach would have a negative impact on both the makespan and the total tardiness. Indeed, as can be seen from Figure 5, for all groups, the super Pareto front moves from left to right and from bottom to top: this indicates higher makespan and higher total tardiness values for all points belonging to the super Pareto front. This can be further noticed from Table 10, which reports the ideal and preferable solutions for all instances, considering that all parts are produced with $SR = 1$, as well as the percentage difference with the ideal and preferable solutions obtained when parts are produced according to their required SR. Indeed, it can be noticed that, for the makespan, the difference is mainly close to 50%, while the total tardiness is majorly affected since, for most of the problems, when parts are printed with $SR = 1$, a total tardiness equal to 0 cannot be achieved anymore. Therefore, considering the required SR when producing parts (as suggested in this work) leads to great advantages in terms of makespan and total tardiness reduction, compared to the situation in which all parts are produced with the highest strength achievable, as AM manufacturers might be tempted to do. Further, according to Table 10, the number of non-dominated points is noticeably higher, compared to the problems reported in Table 9 (with the required customers' strength ratio). Finally, also for these problems the makespan trade-off is noticeably lower than the total tardiness trade-off (Table 10), which implies that selecting a point close to the extreme point with the minimum total tardiness is always preferable, as it leads to a lower distance from the ideal point. Considering the machine features, we can note that problems with bigger machines achieve lower values of the makespan and the total tardiness. In case the required SR is not known in advance, these types of machines should be preferred. Nevertheless, there is a negative impact on costs, given that larger machines imply higher costs.

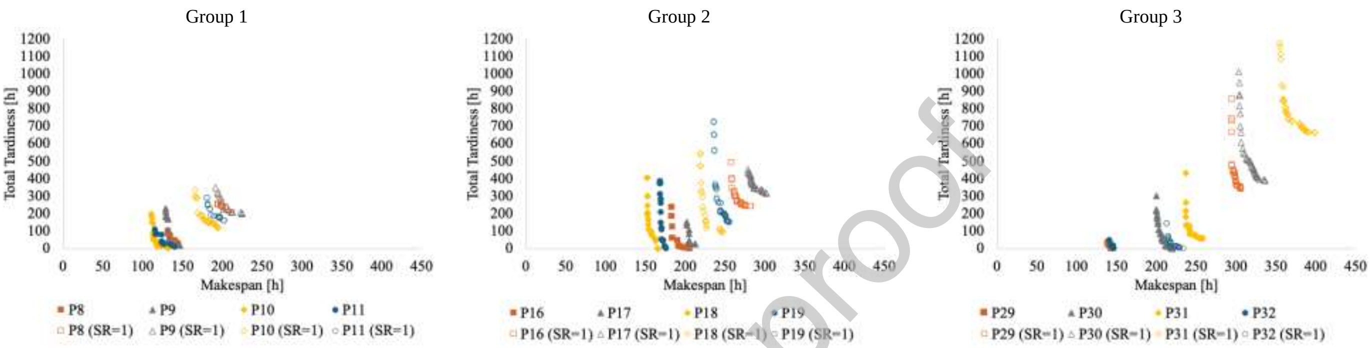

Figure 5. Comparison of the super Pareto front for parts printed with their required SR and with $SR = 1$.

Table 10. Analysis of the ideal and preferable solutions for SR=1.

Group	# Parts	Problem ID	# Machines	# Points Super Pareto Front	Makespan trade-off	Total Tardiness trade-off	Ideal solution ($SR = 1$)		Preferable solution ($SR = 1$)			Ideal solution difference		Preferable solution difference	
							Makespan	Total tardiness	Distance	Makespan	Total tardiness	Makespan	Total tardiness	Makespan	Total tardiness
1	14	P8	2	19	9.18%	57.11%	194.651	200.083	0.091	212.350	200.475	47.61%	1104.01%	45.93%	1106.37%
		P9		14	17.19%	77.31%	191.982	197.121	0.114	212.524	204.962	48.69%	1080.44%	46.06%	1083.45%
		P10	3	34	17.26%	187.10%	166.348	116.934	0.172	195.055	116.936	49.01%	N/D	47.47%	N/D
		P11		20	11.75%	83.85%	181.482	157.947	0.117	202.633	158.285	56.09%	2014.13%	43.70%	2018.66%
2	20	P16	2	31	9.46%	105.92%	258.188	239.843	0.066	274.431	244.915	41.14%	N/D	33.72%	N/D
		P17		38	8.05%	43.72%	278.634	314.069	0.075	297.723	323.637	38.19%	1277.56%	40.11%	1319.52%
		P18	3	24	12.96%	469.01%	218.500	95.325	0.124	244.918	97.834	43.78%	N/D	48.15%	N/D
		P19		46	8.45%	380.08%	235.503	150.935	0.077	252.850	154.128	39.85%	N/D	43.83%	N/D
3	30	P29	3	28	3.85%	150.96%	294.968	340.300	0.038	306.198	341.177	110.04%	N/D	111.86%	N/D
		P30		37	11.08%	159.75%	303.561	389.358	0.086	327.909	401.237	51.61%	N/D	48.80%	N/D
		P31		37	12.43%	77.88%	355.143	661.865	0.089	383.054	689.444	49.59%	1138.01%	49.32%	1195.48%
		P32		37	9.82%	N/D	213.571	0	1.005	234.556	0	50.49%	N/D	59.69%	N/D

7. Conclusions and future work

Additive manufacturing is attracting increasing attention from both academia and industry, especially within Maas businesses. When scheduling their production, these businesses face opposing objectives. On the one hand, they wish to print as many parts as possible within each batch to minimize the makespan, maximizing AM machines' usage rate. On the other hand, however, such an approach would reduce the flexibility required to minimize the tardiness, thus ensuring the respect of customers' due dates. The literature dealing with such a bi-objective problem is lacking, and this work aims to fill the existing research gap by developing a mixed-integer linear programming model that minimizes both the makespan and the total tardiness. In doing so, a parallel unrelated AM batch scheduling problem is considered, where metallic parts can be produced via different SLM machines. Moreover, for the first time in the literature, we consider that customers not only have requirements in terms of due dates, but also in terms of the strength that the parts should possess: if the parts are characterized by a strength lower than what was requested, these would fail unexpectedly and prematurely, with detrimental consequences for safety and costs. As the achievable strength is affected by the orientation with which parts are produced, this work limits the possible orientations with which the parts can be produced to ensure that they are produced with the required strength.

Due to the complexity of the problem and its NP-hard nature, an ε-constraint algorithm and an NSGA-II is proposed to obtain the Pareto front, especially when the problem size becomes larger. More specifically, to improve the algorithm's search capability and solution-building performance, specific decoding mechanisms called the *guided dispatch rule* and the *stochastic batch opening policy* have been developed and integrated into the NSGA-II. Four variants of the NSGA-II, altering the combinatorial inclusion of these mechanisms, have been developed, and their efficiencies have been investigated under pairwise comparisons through computational tests. Discussions have been provided regarding the influence of the integrated mechanisms on the performance of the NSGA-II, concluding that they improve the algorithm's search capacity. It has also been observed that the variants using the guided dispatch rule (NSGA II-GF and NSGA II-GO) are superior to others in terms of minimizing the objective values. Furthermore, through an in-depth analysis of specific numerical instances, we have derived some useful managerial insights regarding the impact of the number of machines and features, and of the strength requirements on the makespan and the total tardiness. Based on our results, we can conclude that machine features (length, width, height) strongly influence both objective functions, and that bigger machines should be preferred if the investment cost does not represent a constraint for the company, as they guarantee lower values of the makespan and the total tardiness. Moreover, we observe how companies should produce parts according to their required strength instead of producing all parts with their maximum strength since the second approach has a considerable negative impact on the makespan and the total tardiness, with both increasing considerably (e.g., the makespan increases up to 50%, and the total tardiness difference is never lower than 1000%, with the exception of cases with the minimum value equal to 0).

As described above, this research represents the first work to consider customer requirements in terms of the desired strength. Being the first work, it is characterized by some limitations. As an example, only three production angles and corresponding achievable strengths are considered; nevertheless, a more continuous distribution of production angles (and corresponding strengths) could be identified from the material science literature and implemented in this work. Notably, the model developed here allows us to do so. Furthermore, this work could be extended and improved by integrating the decoding mechanisms proposed in this paper into other multi-objective meta-heuristics for performance improvement. As an example, energy- and cost-oriented objective functions could be considered as additional objectives to the model proposed in this work. Indeed, due to the high cost of AM machines, the cost-related objective function should be included; this could also be used to guide practitioners in understanding which (and how many) AM machines to buy. Then, AM is a very energy-intensive process (Mecheter et al., 2023), with an impact on both energy costs and emissions (Gaggero et al., 2024); hence, this aspect should also be included. Furthermore, multi-site AM plants (Zehetner & Gansterer, 2022) could be subject to consideration to minimize environmental harms, such as carbon emissions (Kucukkoc, 2023) considering different machine specifications. This can also incorporate irregular shaped parts (Nascimento et al., 2024), and sequence dependent setup times between batches (Yepes-Borrero, 2021). Finally, the model used in this article could be extended to consider the use of different materials, which can also have an effect on different mechanical properties of the parts produced.

Supplementary materials

Supplementary material can be found here: https://github.com/XY122492/AM-Scheduling (Folder “Supplementary Material 1” contains test data used for computational tests, folder “Supplementary Material 2” contains the super Pareto solutions over all algorithms, folder “Supplementary Material 3” contains the statistical test results.

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Gulsen Aydin Keskin: *Conceptualization, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing.*

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